Head ROM During Five Activities of Daily Living

A Single Participant Smartphone IMU Feasibility Study

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Abstract

I tested a forehead-mounted smartphone for recording head range of motion (ROM) during five daily activities. One participant completed three trials per task. MATLAB code processes fixed 180 s windows and compares trial-level ROM. Mean flexion-extension ROM ranged from 9.7° to 19.5°, with a coefficient of variation (CV) of 21–46%. The setup recorded task-related movement, but accuracy remains unverified and the results are specific to this participant.

1 Introduction

Daily activities require different amounts of head movement [1]. These demands matter when designing head-support devices [2]. Studies have assessed wearable sensors for head orientation [3,4], but their validation does not establish the accuracy of this phone setup.

My aim was to record five simple activities without laboratory equipment and compare head ROM across repeated trials.

2 Methodology

A phone was centred on the forehead, held by a cap strap between the phone and cover. I recorded seated laptop work, standing laptop work, standing phone use, eating, and level walking three times each, plus one active calibration. Walking followed straight indoor laps with approximately 180° changes in direction.

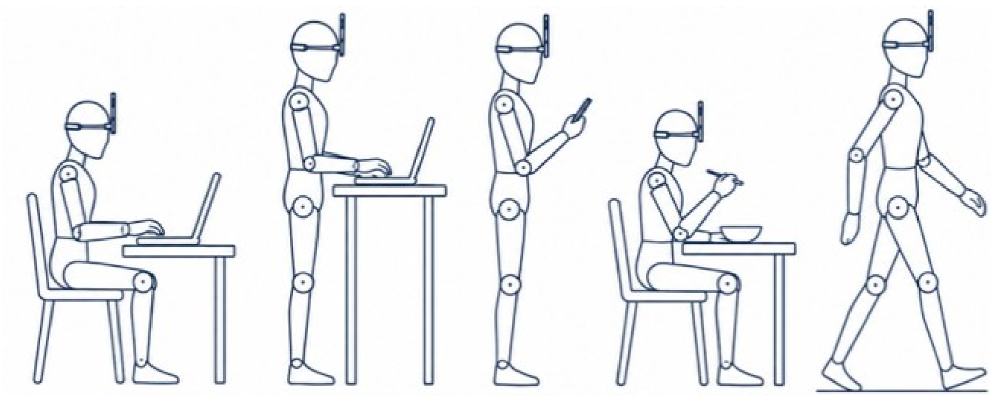

Figure 1 Task positions, not to scale. Seated laptop: 54 cm away, screen top 18 cm below eye level. Standing laptop: 62 cm away, screen top 12 cm above eye level.

Gyroscope and acceleration samples were linearly interpolated onto a common 50 Hz time grid using recorded timestamps. The code selects the quietest complete 10 s block within the first 30 s as the neutral reference. Each 180 s analysis window starts 2 s after that block. Second-order Butterworth filters are applied forward and backward: 0.5 Hz for gravity direction and 5 Hz for yaw rate. Gravity separation has literature precedent [5]; a related head-motion study used 6 Hz [6]. These are processing choices, not validated settings for this phone. Phone axes are mapped to head directions [7]. Flexion and lateral angles come from gravity direction; axial angle comes from bias-corrected yaw-rate integration with a fitted linear trend removed. Walking samples exceeding 25°/s in absolute yaw rate, plus the surrounding 2 s moving window, are excluded. This threshold is a study choice; related turning work used a 30°/s condition [8].

Task ROM is the 97.5th minus the 2.5th percentile of each retained angle channel. Calibration ROM uses maximum minus minimum. Mean and sample standard deviation (SD) are calculated across three trial ROM values for each valid plane. Flexion-extension variance is SD squared; CV is 100 × SD / mean [9]. These statistics describe trial-to-trial spread, not measurement accuracy.

3 Results

All 15 task recordings were processed. The measured sample rate averaged 49.92 Hz and calibration ROM was 119.6°. Standing phone use had the largest mean flexion-extension ROM; eating had the lowest CV (Table 1). Walking retained 77% of its analysis windows after turn exclusion.

Table 1 Head ROM across three trials per task. Variance and CV refer to flexion-extension. NR means not reported; active range is a percentage of calibration ROM.

Task	Flex ext mean ± SD (deg)	Variance (deg2)	CV (%)	Lateral mean ± SD (deg)	Axial mean ± SD (deg)	Active range (%)
Seated laptop	10.8 ± 4.2	17.7	39	5.2 ± 2.3	6.9 ± 3.8	9
Standing laptop	9.7 ± 3.4	11.7	35	3.5 ± 1.2	6.0 ± 1.4	8
Standing phone	19.5 ± 8.9	79.6	46	3.8 ± 1.0	6.8 ± 3.4	16
Eating	13.1 ± 2.7	7.5	21	5.4 ± 2.6	15.0 ± 12.6	11
Walking	11.7 ± 4.4	19.1	37	3.6 ± 0.6	NR	10

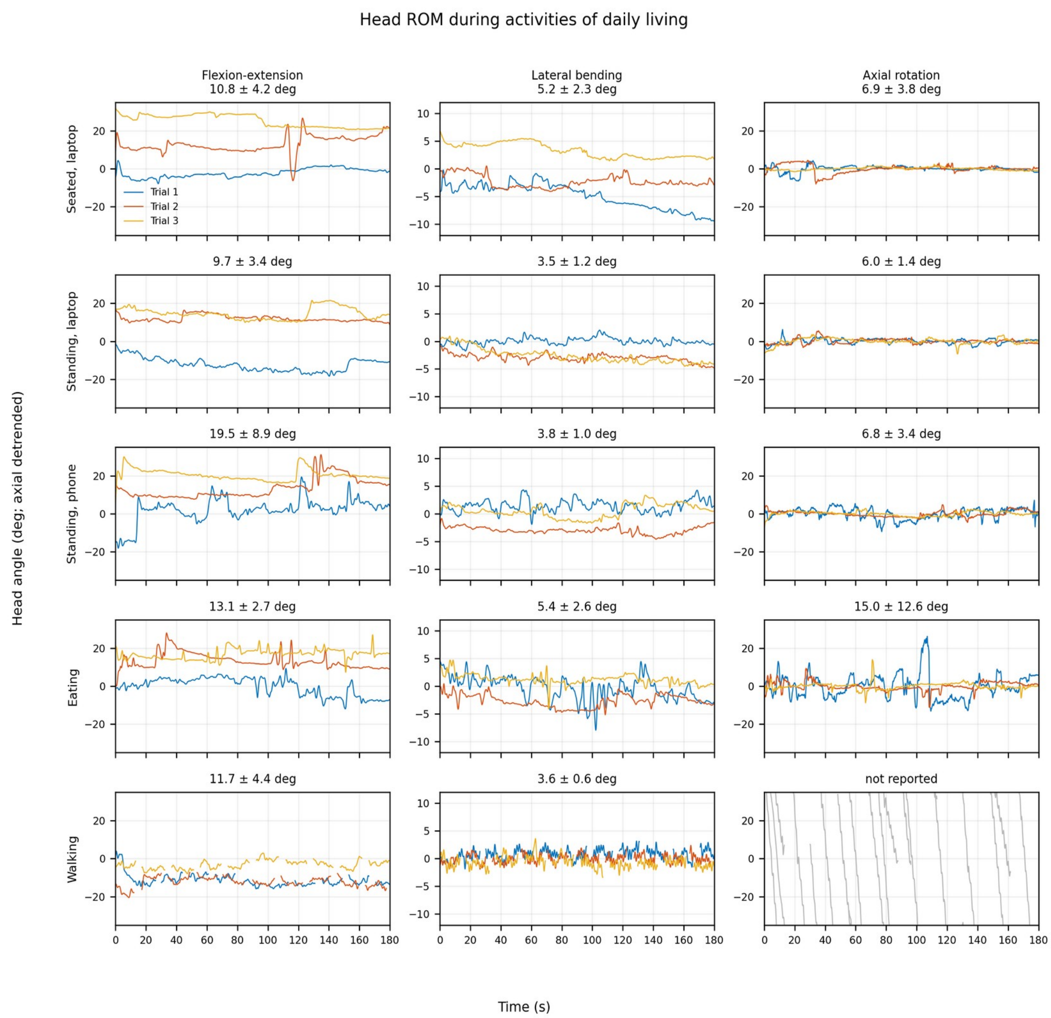

Figure 2 Three trials per task, with time measured from the analysis-window start. Flexion and lateral angles use the neutral reference; axial angles are detrended. Gaps mark excluded walking turns. Walking axial traces are diagnostic only.

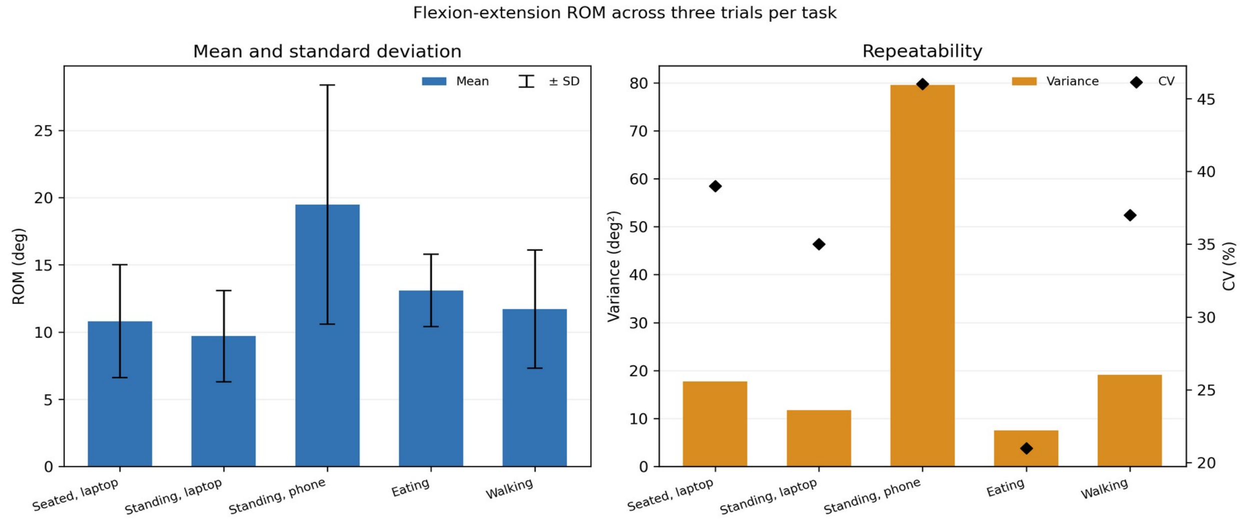

Figure 3 Flexion-extension ROM mean ± sample SD, variance, and CV across three trials per task.

4 Discussion

CV of 21–46% shows that trial ROM varied substantially relative to each task mean. Unscripted movement and mount refitting may explain part of this spread, but the study cannot separate their effects from sensor error.

Neutral-hold SD averaged 1.12°. It describes movement during the selected hold, including possible sensor noise and strap movement; it is not an accuracy estimate. A quiet interval also does not prove the head was in a neutral posture.

At turn thresholds of 20, 25, and 30°/s, mean walking flexion-extension ROM was 11.94°, 11.69°, and 11.68°, with 69.27%, 76.53%, and 77.37% retained. Nearby threshold choices changed data retention more than this ROM result. These sensitivity checks are stored separately from the main analysis.

The next recordings need a repeatable mount position, a fixed task script, and a trunk sensor. A reference instrument and more participants are needed to assess accuracy and whether the findings extend beyond this participant.

5 Limitations

This study has one participant and three trials per task, with no reference measurement. A head-mounted phone measures orientation in space, including trunk motion. The 0.5 Hz gravity filter attenuates faster movement, so flexion and lateral ROM describe slow inclination. Yaw integration can accumulate error, while detrending can remove genuine slow rotation. Walking axial ROM is excluded because lap turns mix head and whole-body rotation. The tables and figures were reproduced independently in Python; the revised MATLAB code still requires a MATLAB runtime check.

6 Conclusion

The setup produced head-motion traces and trial-level ROM for five daily activities. Mean flexion-extension ROM was 9.7–19.5°, with CV of 21–46%. It is a feasible recording and analysis demonstration, but it does not yet provide validated reference values.

References

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